

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I Pages 2–6

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 7–17

90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 What are the values of a , b and c for which the following identity is true?

$$\frac{5x^2 - x + 1}{x(x^2 + 1)} = \frac{a}{x} + \frac{bx + c}{x^2 + 1}$$

- (A) $a = 1, b = 6, c = 1$
 - (B) $a = 1, b = 4, c = 1$
 - (C) $a = 1, b = 6, c = -1$
 - (D) $a = 1, b = 4, c = -1$
- 2 The polynomial $P(z)$ has real coefficients, and $z = 2 - i$ is a root of $P(z)$.

Which quadratic polynomial must be a factor of $P(z)$?

- (A) $z^2 - 4z + 5$
 - (B) $z^2 + 4z + 5$
 - (C) $z^2 - 4z + 3$
 - (D) $z^2 + 4z + 3$
- 3 What is the eccentricity of the ellipse $9x^2 + 16y^2 = 25$?

- (A) $\frac{7}{16}$
- (B) $\frac{\sqrt{7}}{4}$
- (C) $\frac{\sqrt{15}}{4}$
- (D) $\frac{5}{4}$

4 Given $z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$, which expression is equal to $(\bar{z})^{-1}$?

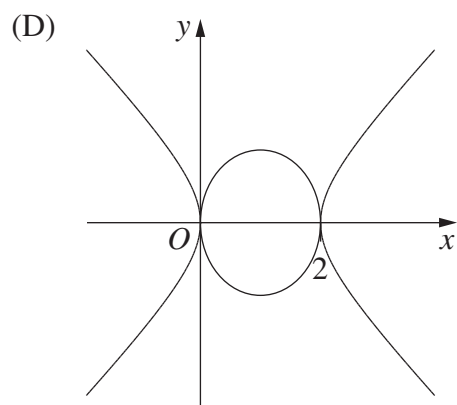
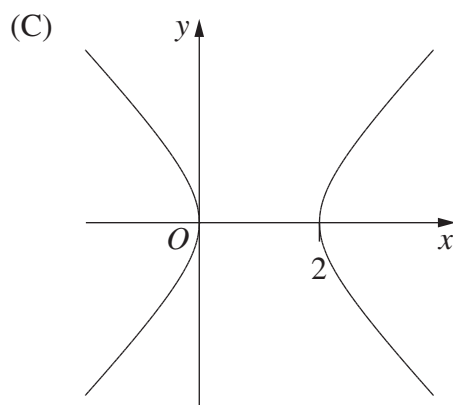
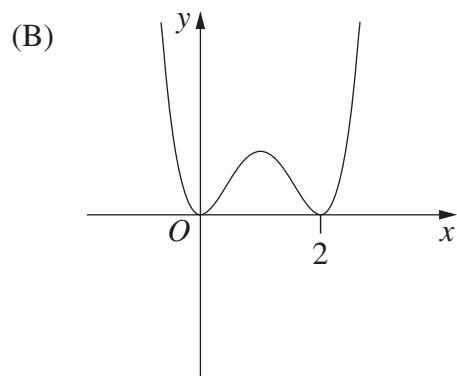
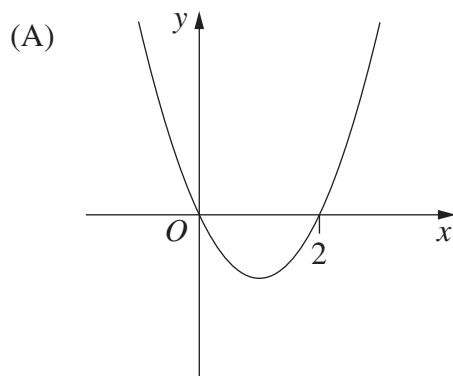
(A) $\frac{1}{2}\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)$

(B) $2\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)$

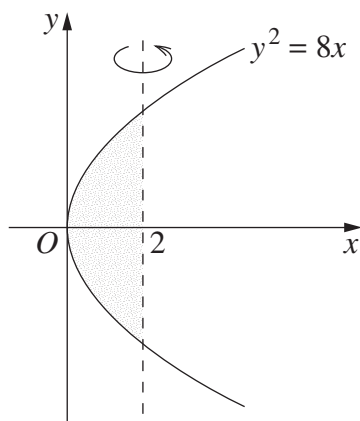
(C) $\frac{1}{2}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

(D) $2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

5 Which graph best represents the curve $y^2 = x^2 - 2x$?



- 6 The region bounded by the curve $y^2 = 8x$ and the line $x = 2$ is rotated about the line $x = 2$ to form a solid.



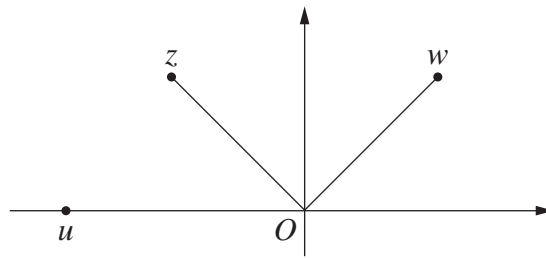
Which expression represents the volume of the solid?

- (A) $\pi \int_0^4 2^2 - \left(\frac{y^2}{8}\right)^2 dy$
- (B) $2\pi \int_0^4 2^2 - \left(\frac{y^2}{8}\right)^2 dy$
- (C) $\pi \int_0^4 \left(2 - \frac{y^2}{8}\right)^2 dy$
- (D) $2\pi \int_0^4 \left(2 - \frac{y^2}{8}\right)^2 dy$

7 Which expression is equal to $\int \frac{1}{1 - \sin x} dx$?

- (A) $\tan x - \sec x + c$
- (B) $\tan x + \sec x + c$
- (C) $\log_e(1 - \sin x) + c$
- (D) $\frac{\log_e(1 - \sin x)}{-\cos x} + c$

8 The Argand diagram shows the complex numbers w , z and u , where w lies in the first quadrant, z lies in the second quadrant and u lies on the negative real axis.



Which statement could be true?

- (A) $u = zw$ and $u = z + w$
- (B) $u = zw$ and $u = z - w$
- (C) $z = uw$ and $u = z + w$
- (D) $z = uw$ and $u = z - w$

9 A particle is moving along a straight line so that initially its displacement is $x = 1$, its velocity is $v = 2$, and its acceleration is $a = 4$.

Which is a possible equation describing the motion of the particle?

- (A) $v = 2\sin(x - 1) + 2$
- (B) $v = 2 + 4\log_e x$
- (C) $v^2 = 4(x^2 - 2)$
- (D) $v = x^2 + 2x + 4$

10 Which integral is necessarily equal to $\int_{-a}^a f(x) dx$?

(A) $\int_0^a f(x) - f(-x) dx$

(B) $\int_0^a f(x) - f(a-x) dx$

(C) $\int_0^a f(x-a) + f(-x) dx$

(D) $\int_0^a f(x-a) + f(a-x) dx$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Consider the complex numbers $z = -2 - 2i$ and $w = 3 + i$.

(i) Express $z + w$ in modulus–argument form. **2**

(ii) Express $\frac{z}{w}$ in the form $x + iy$, where x and y are real numbers. **2**

(b) Evaluate $\int_0^{\frac{1}{2}} (3x - 1) \cos(\pi x) dx$. **3**

(c) Sketch the region in the Argand diagram where $|z| \leq |z - 2|$ and $-\frac{\pi}{4} \leq \arg z \leq \frac{\pi}{4}$. **3**

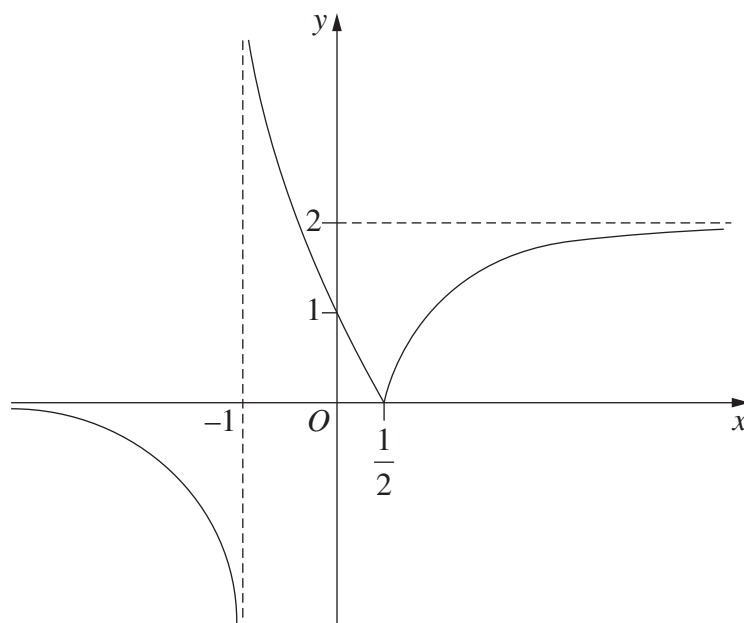
(d) Without the use of calculus, sketch the graph $y = x^2 - \frac{1}{x^2}$, showing all intercepts. **2**

(e) The region enclosed by the curve $x = y(6 - y)$ and the y -axis is rotated about the x -axis to form a solid. **3**

Using the method of cylindrical shells, or otherwise, find the volume of the solid.

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) The diagram shows the graph of a function $f(x)$.



Draw a separate half-page graph for each of the following functions, showing all asymptotes and intercepts.

(i) $y = f(|x|)$ **2**

(ii) $y = \frac{1}{f(x)}$ **2**

- (b) It can be shown that $4\cos^3\theta - 3\cos\theta = \cos 3\theta$. (Do NOT prove this.)

Assume that $x = 2\cos\theta$ is a solution of $x^3 - 3x = \sqrt{3}$.

(i) Show that $\cos 3\theta = \frac{\sqrt{3}}{2}$. **1**

(ii) Hence, or otherwise, find the three real solutions of $x^3 - 3x = \sqrt{3}$. **2**

Question 12 continues on page 9

Question 12 (continued)

- (c) The point $P(x_0, y_0)$ lies on the curves $x^2 - y^2 = 5$ and $xy = 6$. **3**

Prove that the tangents to these curves at P are perpendicular to one another.

- (d) Let $I_n = \int_0^1 \frac{x^{2n}}{x^2 + 1} dx$, where n is an integer and $n \geq 0$.

- (i) Show that $I_0 = \frac{\pi}{4}$. **1**

- (ii) Show that $I_n + I_{n-1} = \frac{1}{2n-1}$. **2**

- (iii) Hence, or otherwise, find $\int_0^1 \frac{x^4}{x^2 + 1} dx$. **2**

End of Question 12

Please turn over

Question 13 (15 marks) Use a SEPARATE writing booklet.

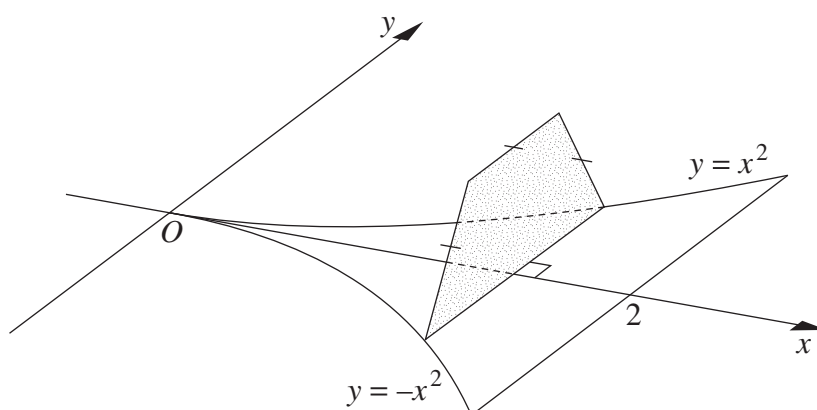
- (a) Using the substitution $t = \tan \frac{x}{2}$, or otherwise, evaluate

3

$$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{1}{3 \sin x - 4 \cos x + 5} dx.$$

- (b) The base of a solid is the region bounded by $y = x^2$, $y = -x^2$ and $x = 2$. Each cross-section perpendicular to the x -axis is a trapezium, as shown in the diagram. The trapezium has three equal sides and its base is twice the length of any one of the equal sides.

4



Find the volume of the solid.

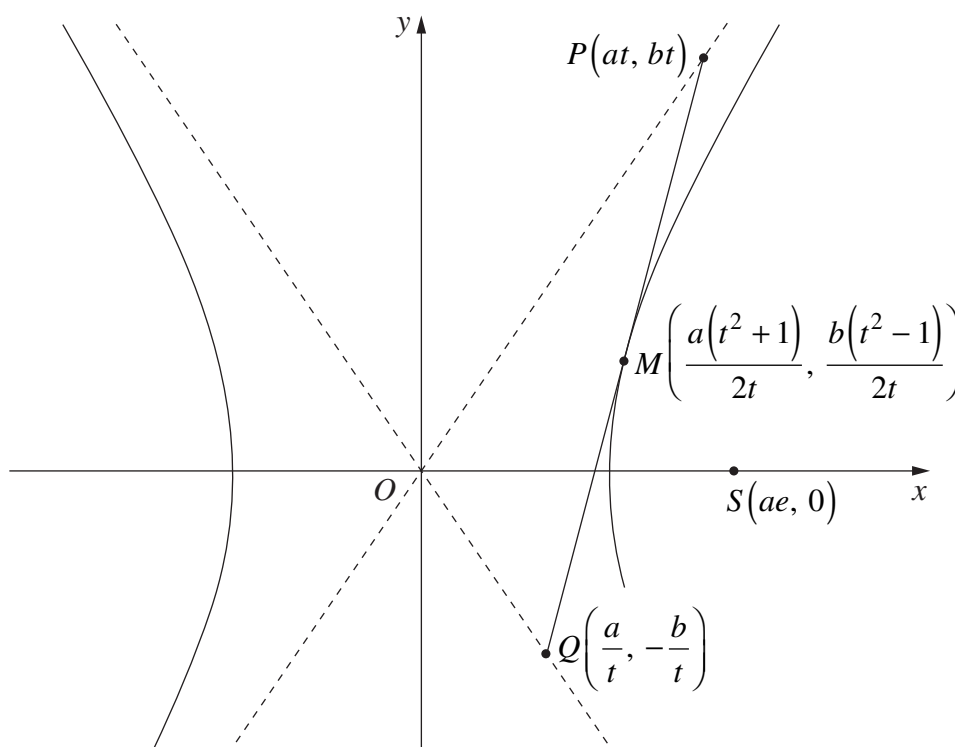
Question 13 continues on page 11

Question 13 (continued)

- (c) The point $S(ae, 0)$ is the focus of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ on the positive x -axis.

The points $P(at, bt)$ and $Q\left(\frac{a}{t}, -\frac{b}{t}\right)$ lie on the asymptotes of the hyperbola, where $t > 0$.

The point $M\left(\frac{a(t^2 + 1)}{2t}, \frac{b(t^2 - 1)}{2t}\right)$ is the midpoint of PQ .



- | | |
|--|---|
| (i) Show that M lies on the hyperbola. | 1 |
| (ii) Prove that the line through P and Q is a tangent to the hyperbola at M . | 3 |
| (iii) Show that $OP \times OQ = OS^2$. | 2 |
| (iv) If P and S have the same x -coordinate, show that MS is parallel to one of the asymptotes of the hyperbola. | 2 |

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

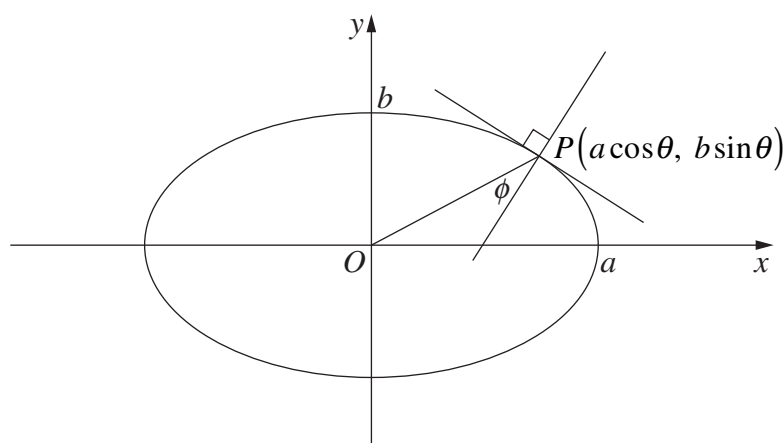
(a) Let $P(x) = x^5 - 10x^2 + 15x - 6$.

(i) Show that $x = 1$ is a root of $P(x)$ of multiplicity three. 2

(ii) Hence, or otherwise, find the two complex roots of $P(x)$. 2

(b) The point $P(a \cos \theta, b \sin \theta)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $a > b$.

The acute angle between OP and the normal to the ellipse at P is ϕ .



(i) Show that $\tan \phi = \left(\frac{a^2 - b^2}{ab} \right) \sin \theta \cos \theta$. 3

(ii) Find a value of θ for which ϕ is a maximum. 2

Question 14 continues on page 13

Question 14 (continued)

- (c) A high speed train of mass m starts from rest and moves along a straight track. At time t hours, the distance travelled by the train from its starting point is x km, and its velocity is v km/h.

The train is driven by a constant force F in the forward direction. The resistive force in the opposite direction is Kv^2 , where K is a positive constant. The terminal velocity of the train is 300 km/h.

- (i) Show that the equation of motion for the train is

2

$$m\ddot{x} = F \left[1 - \left(\frac{v}{300} \right)^2 \right].$$

- (ii) Find, in terms of F and m , the time it takes the train to reach a velocity of 200 km/h.

4

End of Question 14

Please turn over

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) Three positive real numbers a , b and c are such that $a + b + c = 1$ and $a \leq b \leq c$. **2**

By considering the expansion of $(a + b + c)^2$, or otherwise, show that

$$5a^2 + 3b^2 + c^2 \leq 1.$$

- (b) (i) Using de Moivre's theorem, or otherwise, show that for every positive integer n , **2**

$$(1 + i)^n + (1 - i)^n = 2(\sqrt{2})^n \cos \frac{n\pi}{4}.$$

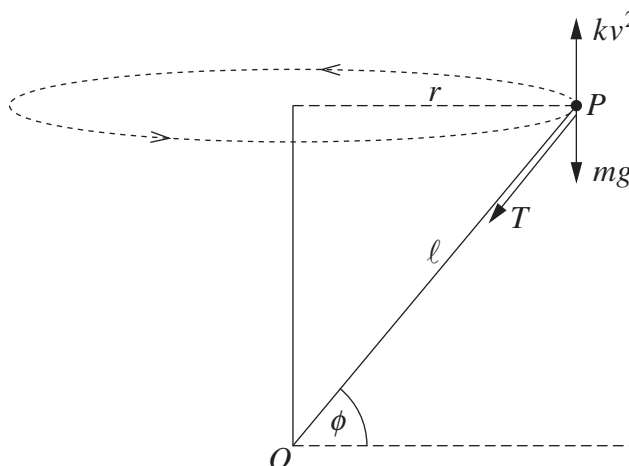
- (ii) Hence, or otherwise, show that for every positive integer n divisible by 4, **3**

$$\binom{n}{0} - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \cdots + \binom{n}{n} = (-1)^{\frac{n}{4}} (\sqrt{2})^n.$$

Question 15 continues on page 15

Question 15 (continued)

- (c) A toy aeroplane P of mass m is attached to a fixed point O by a string of length ℓ . The string makes an angle ϕ with the horizontal. The aeroplane moves in uniform circular motion with velocity v in a circle of radius r in a horizontal plane.



The forces acting on the aeroplane are the gravitational force mg , the tension force T in the string and a vertical lifting force kv^2 , where k is a positive constant.

- (i) By resolving the forces on the aeroplane in the horizontal and the vertical directions, show that $\frac{\sin \phi}{\cos^2 \phi} = \frac{\ell k}{m} - \frac{\ell g}{v^2}$. 3

- (ii) Part (i) implies that $\frac{\sin \phi}{\cos^2 \phi} < \frac{\ell k}{m}$. (Do NOT prove this.) 2

Use this to show that

$$\sin \phi < \frac{\sqrt{m^2 + 4\ell^2 k^2} - m}{2\ell k}.$$

- (iii) Show that $\frac{\sin \phi}{\cos^2 \phi}$ is an increasing function of ϕ for $-\frac{\pi}{2} < \phi < \frac{\pi}{2}$. 2

- (iv) Explain why ϕ increases as v increases. 1

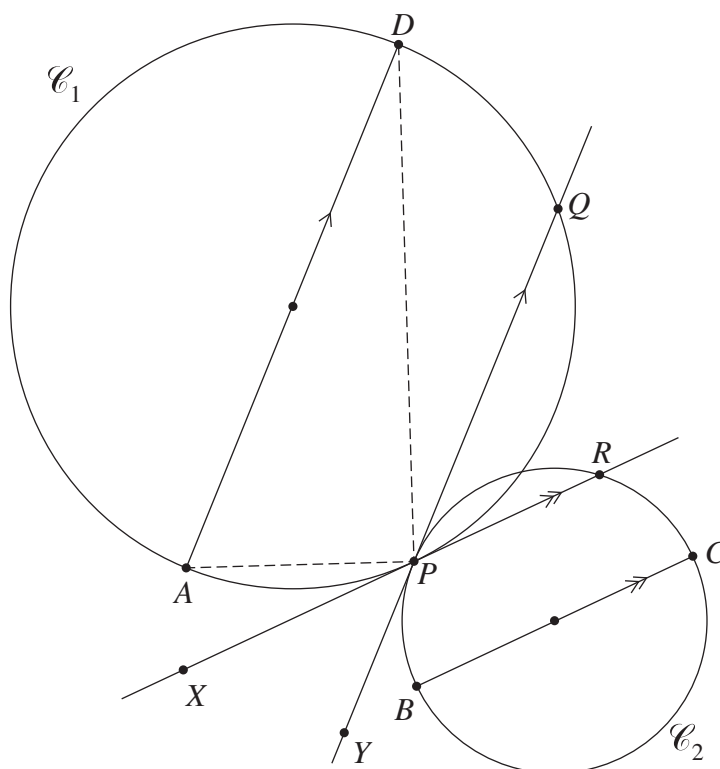
End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) The diagram shows two circles $\mathcal{C}_1$ and $\mathcal{C}_2$. The point P is one of their points of intersection. The tangent to $\mathcal{C}_2$ at P meets $\mathcal{C}_1$ at Q , and the tangent to $\mathcal{C}_1$ at P meets $\mathcal{C}_2$ at R .

The points A and D are chosen on $\mathcal{C}_1$ so that AD is a diameter of $\mathcal{C}_1$ and parallel to PQ . Likewise, points B and C are chosen on $\mathcal{C}_2$ so that BC is a diameter of $\mathcal{C}_2$ and parallel to PR .

The points X and Y lie on the tangents PR and PQ , respectively, as shown in the diagram.



Copy or trace the diagram into your writing booklet.

- | | |
|---|---|
| (i) Show that $\angle APX = \angle DPQ$. | 2 |
| (ii) Show that A , P and C are collinear. | 3 |
| (iii) Show that $ABCD$ is a cyclic quadrilateral. | 1 |

Question 16 continues on page 17

Question 16 (continued)

(b) Suppose n is a positive integer.

(i) Show that **3**

$$-x^{2n} \leq \frac{1}{1+x^2} - \left(1 - x^2 + x^4 - x^6 + \cdots + (-1)^{n-1} x^{2n-2}\right) \leq x^{2n}.$$

(ii) Use integration to deduce that **2**

$$-\frac{1}{2n+1} \leq \frac{\pi}{4} - \left(1 - \frac{1}{3} + \frac{1}{5} - \cdots + (-1)^{n-1} \frac{1}{2n-1}\right) \leq \frac{1}{2n+1}.$$

(iii) Explain why $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$. **1**

(c) Find $\int \frac{\ln x}{(1 + \ln x)^2} dx$. **3**

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

2014 HSC Mathematics Extension 2 Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	D
2	A
3	B
4	C
5	C
6	D
7	B
8	B
9	A
10	D

Section II

Question 11 (a) (i)

Criteria	Marks
• Provides a correct solution	2
• Finds modulus or argument, or equivalent merit	1

Sample answer:

$$z + w = (-2 - 2i) + 3 + i = 1 - i$$

$$|z + w| = \sqrt{2}, \quad \arg(z + w) = \frac{-\pi}{4}$$

$$\text{so } z + w = \sqrt{2} \left[\cos\left(\frac{-\pi}{4}\right) + i \sin\left(\frac{-\pi}{4}\right) \right]$$

Question 11 (a) (ii)

Criteria	Marks
• Provides a correct solution	2
• Identifies conjugate, or equivalent merit	1

Sample answer:

$$\frac{z}{w} = \frac{-2 - 2i}{3 + i} \times \frac{3 - i}{3 - i}$$

$$= \frac{-8 - 4i}{10}$$

$$= \frac{-4 - 2i}{5}$$

$$= \frac{-4}{5} - \frac{2}{5}i$$

Question 11 (b)

Criteria	Marks
• Provides a correct solution	3
• Correctly uses limits in one of two terms, or equivalent merit	2
• Attempts integration by parts, or equivalent merit	1

Sample answer:

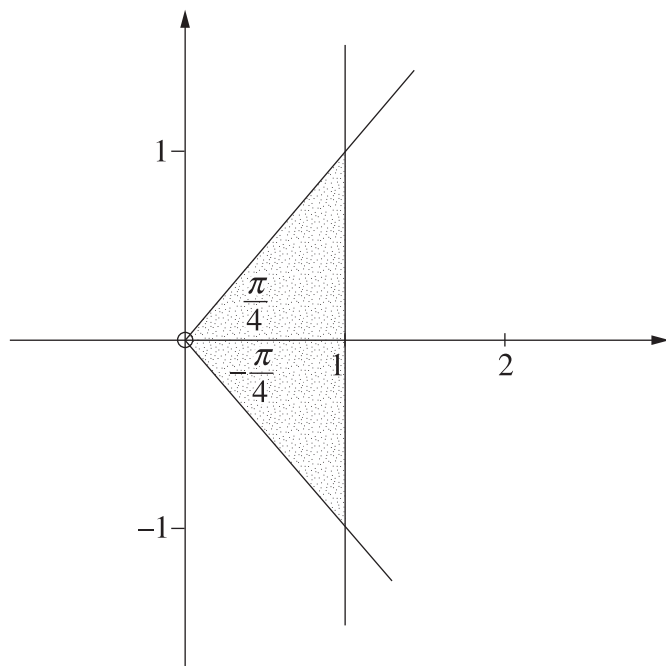
$$\int_0^{\frac{1}{2}} (3x-1)\cos \pi x \, dx = \left[(3x-1)\frac{\sin \pi x}{\pi} \right]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} \frac{3}{\pi} \sin \pi x \, dx$$

$$= \frac{1}{2\pi} - \left[\frac{-3}{\pi^2} \cos \pi x \right]_0^{\frac{1}{2}}$$

$$= \frac{1}{2\pi} - \frac{3}{\pi^2}$$

Question 11 (c)

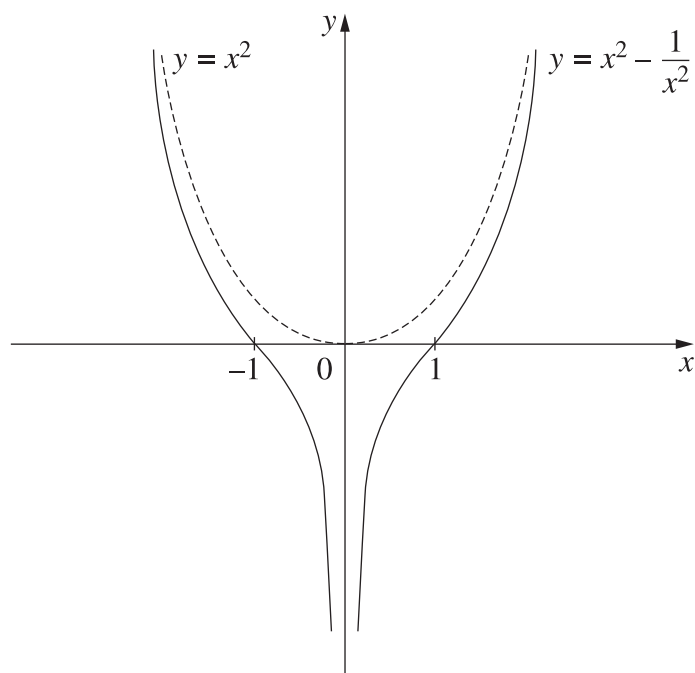
Criteria	Marks
• Draws a correct sketch	3
• Correctly shows region satisfying $ z \leq z - 2 $, or equivalent merit	2
• Correctly shows region satisfying $-\frac{\pi}{4} \leq \arg(z) \leq \frac{\pi}{4}$, or equivalent merit	1

Sample answer:

Question 11 (d)

Criteria	Marks
• Draws correct sketch	2
• Sketches a graph with correct x -intercepts, or equivalent merit	1

Sample answer:

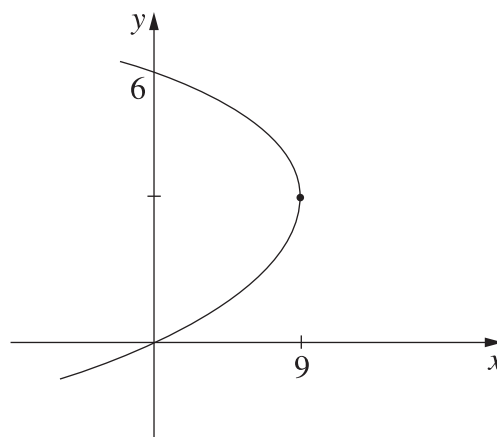


Question 11 (e)

Criteria	Marks
• Provides a correct solution	3
• Attempts to evaluate the integral for the volume, or equivalent merit	2
• Applies a formula for the volume by cylindrical shells, or equivalent merit	1

Sample answer:

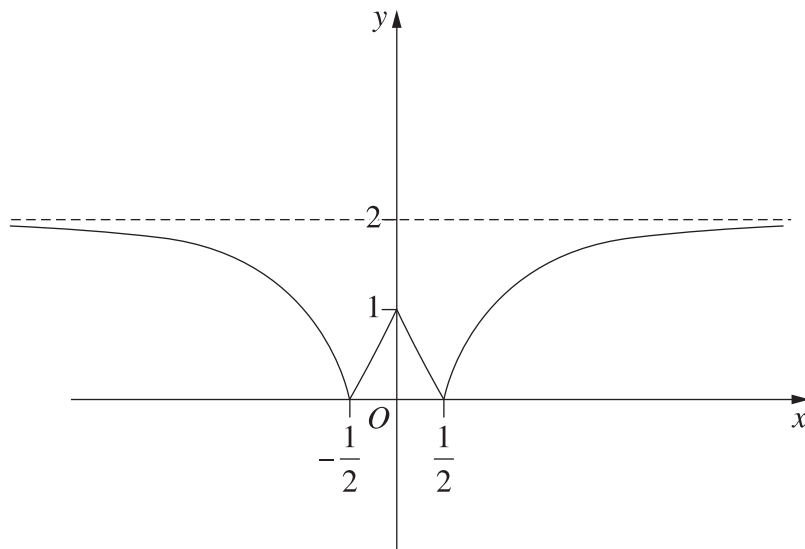
$$\begin{aligned}
 V &= 2\pi \int_0^6 xy \, dy \\
 &= 2\pi \int_0^6 y(6-y)y \, dy \\
 &= 2\pi \int_0^6 (6y^2 - y^3) \, dy \\
 &= 2\pi \left[2y^3 - \frac{y^4}{4} \right]_0^6 \\
 &= 2\pi(2 \times 216 - 36 \times 9)
 \end{aligned}$$



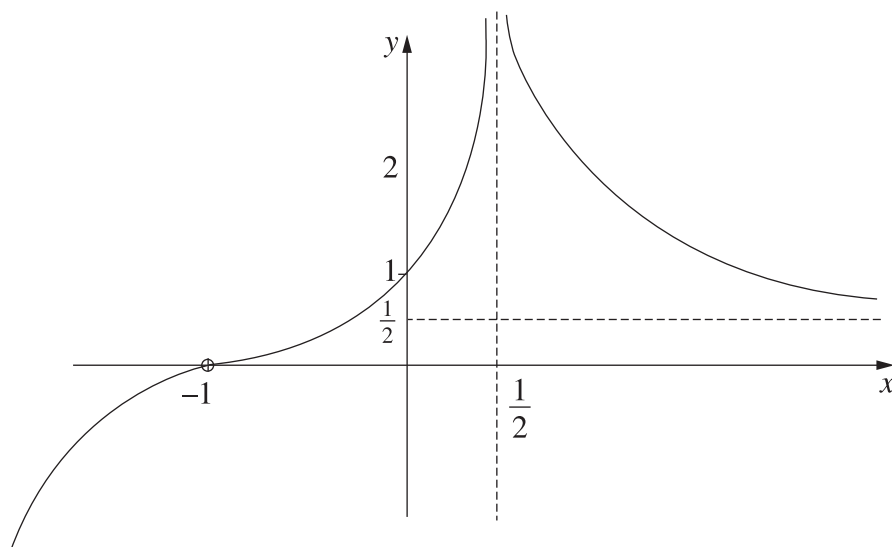
$$\text{Volume} = 216\pi \text{ units}^3$$

Question 12 (a) (i)

Criteria	Marks
• Draws a correct sketch	2
• Sketches graph symmetric with respect to y-axis or equivalent merit	1

Sample answer:**Question 12 (a) (ii)**

Criteria	Marks
• Draws a correct sketch	2
• Shows asymptote at $x = \frac{1}{2}$, or $y = \frac{1}{2}$ or equivalent merit	1

Sample answer:

Question 12 (b) (i)

Criteria	Marks
• Provides a correct solution	1

Sample answer:Substituting $x = 2\cos\theta$

$$8\cos^3\theta - 6\cos\theta = \sqrt{3}$$

$$\text{so } 4\cos^3\theta - 3\cos\theta = \frac{\sqrt{3}}{2}$$

$$\therefore \cos 3\theta = \frac{\sqrt{3}}{2}$$

Question 12 (b) (ii)

Criteria	Marks
• Provides a correct solution	2
• Finds one real solution, or equivalent merit	1

Sample answer:

$$\cos 3\theta = \frac{\sqrt{3}}{2}$$

$$\text{so } 3\theta = \frac{\pi}{6}, \frac{11\pi}{6}, \frac{12\pi}{6}$$

$$\therefore \theta = \frac{\pi}{18}, \frac{11\pi}{18}, \frac{13\pi}{18}$$

Solutions are

$$x = 2\cos\frac{\pi}{18}, 2\cos\frac{11\pi}{18}, 2\cos\frac{13\pi}{18}$$

Question 12 (c)

Criteria	Marks
• Provides a correct solution	3
• Finds the slope of one curve, and attempts to find slope of other curve, or equivalent merit	2
• Attempts to find slope at (x_0, y_0) of one curve using implicit differentiation, or equivalent merit	1

Sample answer:

$$x^2 - y^2 = 5$$

Differentiating with respect to x :

$$2x - 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{x}{y}$$

at point (x_0, y_0) slope is $m_1 = \frac{x_0}{y_0}$

$$xy = 6 \quad (x, y \neq 0)$$

Differentiating with respect to x :

$$y + x \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

at point (x_0, y_0) slope is $m_2 = -\frac{y_0}{x_0}$ If curves meet then $x_0, y_0 \neq 0$

$$\text{and } m_1 \times m_2 = \frac{x_0}{y_0} \times \frac{-y_0}{x_0} = -1$$

Then tangents are perpendicular.

Question 12 (d) (i)

Criteria	Marks
• Provides a correct solution	1

Sample answer:

$$I_0 = \int_0^1 \frac{1}{x^2 + 1} dx$$

$$= \left[\tan^{-1} x \right]_0^1$$

$$= \frac{\pi}{4}$$

Question 12 (d) (ii)

Criteria	Marks
• Provides a correct solution	2
• Writes the sum as a single integral, or equivalent merit	1

Sample answer:

$$I_n + I_{n-1} = \int_0^1 \left(\frac{x^{2n}}{x^2 + 1} + \frac{x^{2n-2}}{x^2 + 1} \right) dx$$

$$= \int_0^1 \frac{x^{2n-2}(x^2 + 1)}{(x^2 + 1)} dx$$

$$= \int_0^1 x^{2n-2} dx$$

$$= \left[\frac{1}{2n-1} x^{2n-1} \right]_0^1$$

$$= \frac{1}{2n-1} (1 - 0)$$

$$= \frac{1}{2n-1}$$

Question 12 (d) (iii)

Criteria	Marks
• Provides a correct answer	2
• Attempts to apply the recursion relation from part (ii), or equivalent merit	1

Sample answer:

$$\int_0^1 \frac{x^4}{x^2+1} dx = I_2$$

$$\text{Now, } I_2 + I_1 = \frac{1}{3}$$

$$I_1 + I_0 = 1$$

Subtracting:

$$I_2 - I_0 = \frac{1}{3} - 1$$

$$= -\frac{2}{3}$$

$$\therefore I_2 = I_0 - \frac{2}{3}$$

$$= \frac{\pi}{4} - \frac{2}{3}$$

Question 13 (a)

Criteria	Marks
• Provides a correct solution	3
• Finds a correct primitive, or equivalent merit	2
• Attempts to obtain an integral in terms of t , or equivalent merit	1

Sample answer:

$$t = \tan \frac{x}{2}, \quad \sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$\begin{aligned} \frac{dt}{dx} &= \frac{1}{2} \sec^2 \frac{x}{2} = \frac{1}{2} \left(1 + \tan^2 \frac{x}{2} \right) \\ &= \frac{1}{2} (1 + t^2) \end{aligned}$$

$$\therefore dx = \frac{2dt}{1+t^2}$$

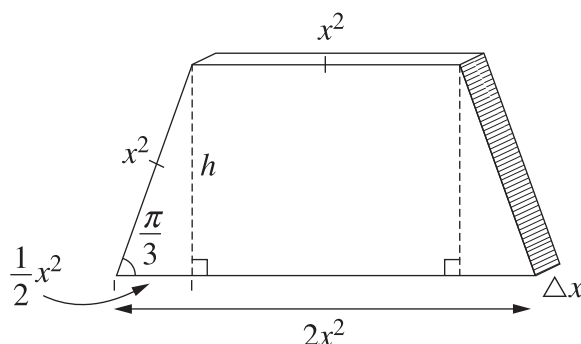
$$\text{when } x = \frac{\pi}{2}, \quad t = 1 \text{ and when } x = \frac{\pi}{3}, \quad t = \frac{1}{\sqrt{3}}$$

Question 13 (a) Sample answer – continued:

$$\begin{aligned}
I &= \int_{\frac{1}{\sqrt{3}}}^1 \frac{2}{3\left(\frac{2t}{1+t^2}\right) - 4\left(\frac{1-t^2}{1+t^2}\right) + 5} \times \frac{1}{(1+t^2)} dt \\
&= \int_{\frac{1}{\sqrt{3}}}^1 \frac{2dt}{6t - 4 + 4t^2 + 5 + 5t^2} \\
&= \int_{\frac{1}{\sqrt{3}}}^1 \frac{2dt}{9t^2 + 6t + 1} \\
&= 2 \int_{\frac{1}{\sqrt{3}}}^1 \frac{dt}{(3t+1)^2} \\
&= -\frac{2}{3} \left[\frac{1}{3t+1} \right]_{\frac{1}{\sqrt{3}}}^1 \\
&= -\frac{2}{3} \left(\frac{1}{4} - \frac{1}{\sqrt{3}+1} \right) \\
&= -\frac{2}{3} \left(\frac{1}{4} - \frac{\sqrt{3}-1}{2} \right) \\
&= -\frac{1}{6} + \frac{\sqrt{3}-1}{3} \\
&= \frac{2\sqrt{3}-3}{6}
\end{aligned}$$

Question 13 (b)

Criteria	Marks
• Provides a correct solution	4
• Finds an expression for the volume of the solid	3
• Finds the area of each cross-section, or equivalent merit	2
• Finds the height of each trapezium, or equivalent merit	1

Sample answer:

$$h = x^2 \sin \frac{\pi}{3} = x^2 \frac{\sqrt{3}}{2}$$

$$\text{Area of cross-section} = \frac{1}{2} \times x^2 \frac{\sqrt{3}}{2} (x^2 + 2x^2)$$

$$= \frac{\sqrt{3}}{4} x^2 \cdot 3x^2$$

$$= \frac{3\sqrt{3}}{4} x^4$$

$$\text{Volume of the slice} = \frac{3\sqrt{3}}{4} x^4 \Delta x$$

$$\therefore \text{Volume of the solid} = \int_0^2 \frac{3\sqrt{3}}{4} x^4 dx$$

$$= \frac{3\sqrt{3}}{4} \left[\frac{x^5}{5} \right]_0^2$$

$$= \frac{3\sqrt{3}}{4 \times 5} (2^5)$$

$$= \frac{24\sqrt{3}}{5} \text{ unit}^3$$

Question 13 (c) (i)

Criteria	Marks
• Provides a correct solution	1

Sample answer:

$$M\left(\frac{a}{2}\left(t + \frac{1}{t}\right), \frac{b}{2}\left(t - \frac{1}{t}\right)\right)$$

Substituting for x and y ,

$$\begin{aligned} \frac{x^2}{a^2} - \frac{y^2}{b^2} &= \frac{1}{4}\left(t + \frac{1}{t}\right)^2 - \frac{1}{4}\left(t - \frac{1}{t}\right)^2 \\ &= \frac{1}{4}\left(t + \cancel{\frac{1}{t}} + t - \cancel{\frac{1}{t}}\right)\left(\cancel{t} + \frac{1}{t} - \cancel{t} + \frac{1}{t}\right) \\ &= \frac{1}{4}2t \cdot \frac{2}{t} \\ &= 1 \end{aligned}$$

So M lies on the hyperbola.

Question 13 (c) (ii)

Criteria	Marks
• Provides a correct proof	3
• Finds the slope of the tangent, or equivalent merit	2
• Finds slope of PQ , or attempts to find the slope of the tangent, or equivalent merit	1

Sample answer:

$$\text{Slope of } PQ \text{ is } \frac{bt + \frac{b}{t}}{at - \frac{a}{t}} = \frac{b\left(t + \frac{1}{t}\right)}{a\left(t - \frac{1}{t}\right)}$$

Differentiating $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with respect to x :

$$\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{xb^2}{ya^2}$$

at M

$$\frac{dy}{dx} = \frac{\frac{a}{2}\left(t + \frac{1}{t}\right)b^2}{\frac{b}{2}\left(t - \frac{1}{t}\right)a^2}$$

$$= \frac{b\left(t + \frac{1}{t}\right)}{a\left(t - \frac{1}{t}\right)}$$

$$= \text{slope of } PQ$$

so PQ is a tangent to hyperbola at M

Question 13 (c) (iii)

Criteria	Marks
• Provides a correct solution	2
• Finds $OP \times OQ$ in terms of a and b , or equivalent merit	1

Sample answer:

$$OP = \sqrt{(a^2 + b^2)t^2}$$

$$OQ = \sqrt{\frac{a^2 + b^2}{t^2}}$$

$$\therefore OP \times OQ = a^2 + b^2$$

$$OS^2 = a^2 e^2$$

$$= a^2 \left(1 + \frac{b^2}{a^2} \right)$$

$$= a^2 + b^2$$

$$= OP \times OQ$$

Question 13 (c) (iv)

Criteria	Marks
• Provides a correct solution	2
• Finds one coordinate of M in terms of a , b and e , or equivalent merit	1

Sample answer:

Equations of asymptotes are $y = \pm \frac{b}{a}x$

If $at = ae$ then $t = e$.

The coords of M are then $\left(\frac{a}{2}\left(e + \frac{1}{e}\right), \frac{b}{2}\left(e - \frac{1}{e}\right)\right)$

slope of MS is

$$\begin{aligned} \frac{\frac{b}{2}\left(e - \frac{1}{e}\right)}{\frac{a}{2}\left(e + \frac{1}{e}\right) - ae} &= \frac{b\left(e - \frac{1}{e}\right)}{ae + \frac{a}{e} - 2ae} \\ &= \frac{b\left(e - \frac{1}{e}\right)}{a\left(\frac{1}{e} - e\right)} \\ &= -\frac{b}{a} \end{aligned}$$

which is the slope of one of the asymptotes and so MS is parallel to it.

Question 14 (a) (i)

Criteria	Marks
• Provides a correct solution	2
• Shows that $P''(1) = 0$, or equivalent merit	1

Sample answer:

$$P'(x) = 5x^4 - 20x + 15$$

$$P''(x) = 20x^3 - 20$$

$$P(1) = 1 - 10 + 15 - 6 = 0$$

$$P'(1) = 5 - 20 + 15 = 0$$

$$P''(1) = 20 - 20 = 0 \quad \therefore x = 1 \text{ is a root of multiplicity 3}$$

Question 14 (a) (ii)

Criteria	Marks
• Provides a correct solution	2
• Finds product and sum of remaining two roots, or equivalent merit	1

Sample answer:Let other roots be α, β

$$\alpha + \beta + 1 + 1 + 1 = 0 \quad , \quad \alpha\beta(1)(1)(1) = 6$$

$$\alpha + \beta = -3 \quad \alpha\beta = 6$$

Hence, α and β are the solutions to $x^2 + 3x + 6 = 0$

$$\begin{aligned} x &= \frac{-3 \pm \sqrt{9 - 4(1)(6)}}{2} \\ &= \frac{-3 \pm \sqrt{-15}}{2} \\ &= \frac{-3 \pm \sqrt{15}i}{2} \end{aligned}$$

The complex roots are $\frac{-3 + \sqrt{15}i}{2}, \frac{-3 - \sqrt{15}i}{2}$

Question 14 (b) (i)

Criteria	Marks
• Provides a correct solution	3
• Uses the slopes of OP and the normal in a correct expression for $\tan \phi$, or equivalent merit	2
• Finds the slope of the tangent at P , or equivalent merit	1

Sample answer:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Differentiating with respect to x :

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{xb^2}{ya^2}$$

$$\text{slope of tangent at } P \text{ is } -\frac{b \cos \theta}{a \sin \theta}$$

$$\text{slope of normal at } P \text{ is } m_1 = \frac{a \sin \theta}{b \cos \theta}$$

$$\text{slope of } OP \text{ is } m_2 = \frac{b \sin \theta}{a \cos \theta}$$

$$\text{Hence, } \tan \phi = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{a \sin \theta}{b \cos \theta} - \frac{b \sin \theta}{a \cos \theta}}{1 + \frac{\sin^2 \theta}{\cos^2 \theta}} \right|$$

$$= \left| \frac{\frac{\sin \theta \left(\frac{a}{b} - \frac{b}{a} \right)}{\cos \theta}}{\frac{\cos^2 \theta + \sin^2 \theta}{\cos^2 \theta}} \right|$$

$$= \frac{\left(\frac{a^2 - b^2}{ab} \right) \frac{\sin \theta}{\cos \theta} \cdot \cos^2 \theta}{\cos^2 \theta + \sin^2 \theta}, a > b$$

$$= \frac{a^2 - b^2}{ab} \sin \theta \cos \theta$$

Question 14 (b) (ii)

Criteria	Marks
• Provides a correct solution	2
• Uses double angle formula, or equivalent merit	1

Sample answer:

$$\tan \phi = \left(\frac{a^2 - b^2}{ab} \right) \frac{\sin 2\theta}{2}$$

as $\sin x$ has maximum at $x = \frac{\pi}{2}$

$\tan \phi$ has maximum at $2\theta = \frac{\pi}{2}$

$\therefore \phi$ is a maximum when $\theta = \frac{\pi}{4}$

Question 14 (c) (i)

Criteria	Marks
• Provides a correct solution	2
• Finds $m\ddot{x} = F - kv^2$, or equivalent merit	1

Sample answer:

From Newton's second law

$$m\ddot{x} = F - kv^2$$

The terminal velocity ($v = 300$) occurs when $\ddot{x} = 0$, so

$$F = k \times (300)^2, \text{ or } k = \frac{F}{(300)^2}$$

$$\begin{aligned} \text{Hence } m\ddot{x} &= F - \frac{F}{(300)^2} v^2 \\ &= F \left(1 - \left(\frac{v}{300} \right)^2 \right) \end{aligned}$$

Question 14 (c) (ii)

Criteria	Marks
• Provides a correct solution	4
• Obtains an expression for t in terms of v , or equivalent merit	3
• Attempts to use partial fractions to find t in terms of v , or equivalent merit	2
• Obtains an equation relating t and v involving integrals, or equivalent merit	1

Sample answer:

$$m \frac{dv}{dt} = F \left(1 - \left(\frac{v}{300} \right)^2 \right)$$

$$\int \frac{dv}{1 - \left(\frac{v}{300} \right)^2} = \int \frac{F}{m} dt$$

$$\frac{F}{m} t = \int \frac{1}{\left(1 - \frac{v}{300} \right) \left(1 + \frac{v}{300} \right)} dv$$

$$= \frac{1}{2} \int \frac{1}{1 - \frac{v}{300}} + \frac{1}{1 + \frac{v}{300}} dv$$

$$= \frac{300}{2} \left[\ln \left(1 + \frac{v}{300} \right) - \ln \left(1 - \frac{v}{300} \right) \right] + c$$

$$v = 0 \text{ when } t = 0 \Rightarrow c = 0$$

$$\text{Thus } t = \frac{m}{F} \times 150 \ln \left(\frac{1 + \frac{v}{300}}{1 - \frac{v}{300}} \right)$$

$$= \frac{150m}{F} \ln \left(\frac{300 + v}{300 - v} \right)$$

$$\text{Put } v = 200, t = \frac{150m}{F} \ln 5$$

Question 15 (a)

Criteria	Marks
• Provides a correct solution	2
• Expands $(a + b + c)^2$ and attempts to use the relative sizes of a, b or c , or equivalent merit	1

Sample answer:

$$a + b + c = 1, \quad a \leq b \leq c$$

$$\text{So } (a + b + c)^2 = 1$$

$$LHS = a^2 + b^2 + c^2 + 2(ab + bc + ac)$$

$$\geq a^2 + b^2 + c^2 + 2(a^2 + b^2 + a^2) \quad (\text{since } b \geq a, c \geq b, c \geq a)$$

$$\therefore LHS \geq 5a^2 + 3b^2 + c^2$$

$$\therefore 5a^2 + 3b^2 + c^2 \leq 1$$

Question 15 (b) (i)

Criteria	Marks
• Provides a correct solution	2
• Writes $1 + i$ or $1 - i$ in modulus argument form, or equivalent merit	1

Sample answer:

Write $(1 + i)$ and $(1 - i)$ in modulus-argument form:

$$\begin{aligned}
 1 + i &= \sqrt{2} \left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \right) & \text{and} & & 1 - i &= \sqrt{2} \left(\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}} \right) \\
 &= \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) & & & &= \sqrt{2} \left(\cos \frac{\pi}{4} - i \sin \frac{\pi}{4} \right) \\
 & & & & &= \sqrt{2} \left(\cos \left(-\frac{\pi}{4} \right) + i \sin \left(-\frac{\pi}{4} \right) \right)
 \end{aligned}$$

Using de Moivre's theorem

$$\begin{aligned}
 (1 + i)^n &= \left[\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \right]^n \\
 &= (\sqrt{2})^n \left(\cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} \right)
 \end{aligned}$$

and

$$\begin{aligned}
 (1 - i)^n &= \left[\sqrt{2} \left(\cos \left(-\frac{\pi}{4} \right) + i \sin \left(-\frac{\pi}{4} \right) \right) \right]^n \\
 &= (\sqrt{2})^n \left(\cos \left(-\frac{n\pi}{4} \right) + i \sin \left(-\frac{n\pi}{4} \right) \right) \\
 &= \sqrt{2}^n \left(\cos \frac{n\pi}{4} - i \sin \frac{n\pi}{4} \right)
 \end{aligned}$$

adding

$$(1 + i)^n + (1 - i)^n = 2(\sqrt{2})^n \cos \frac{n\pi}{4}$$

Question 15 (b) (ii)

Criteria	Marks
• Provides a correct solution	3
• Attempts to add two binomial expressions, or equivalent merit	2
• Applies the binomial theorem to $(1 + i)^n$ or $(1 - i)^n$, or equivalent merit	1

Sample answer:

Using the binomial theorem

$$\begin{aligned}
 (1 + i)^n &= \sum_{k=0}^n \binom{n}{k} i^k \\
 &= \binom{n}{0} + \binom{n}{1} i - \binom{n}{2} - \binom{n}{3} i + \binom{n}{4} + \dots + \binom{n}{n}
 \end{aligned}$$

and

$$\begin{aligned}
 (1 - i)^n &= \sum_{k=0}^n (-1)^k \binom{n}{k} i^k \\
 &= \binom{n}{0} - \binom{n}{1} i - \binom{n}{2} + \binom{n}{3} i + \binom{n}{4} - \dots + \binom{n}{n}
 \end{aligned}$$

The last terms are positive since n is divisible by 4

Adding the identities

$$\begin{aligned}
 (1 + i)^n + (1 - i)^n &= 2 \left[\binom{n}{0} - \binom{n}{2} + \binom{n}{4} - \dots + \binom{n}{n} \right]
 \end{aligned}$$

Combining this with the expressions from (i) for $(1 + i)^n + (1 - i)^n$

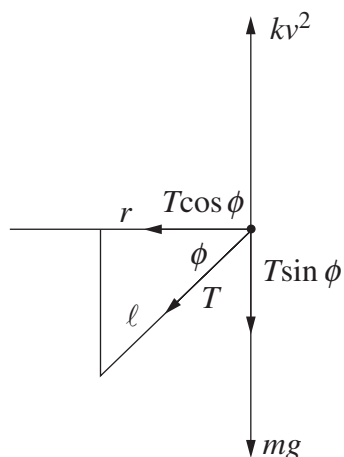
$$\begin{aligned}
 \binom{n}{0} - \binom{n}{2} + \dots + \binom{n}{n} &= (\sqrt{2})^n \cos \frac{n\pi}{4} \\
 &= \sqrt{2}^n (-1)^{\frac{n}{4}}
 \end{aligned}$$

since $\cos \frac{n\pi}{4} = -1$ for odd multiples of $\pi \left(\frac{n}{4} \text{ odd} \right)$

$= 1$ for even multiples of $\pi \left(\frac{n}{4} \text{ even} \right)$

Question 15 (c) (i)

Criteria	Marks
• Provides a correct solution	3
• Resolves forces in both directions and eliminates r , or equivalent merit	2
• Resolves forces in one direction, or equivalent merit	1

Sample answer:

Vertical $kv^2 = T \sin \phi + mg$

$$T \sin \phi = kv^2 - mg$$

Horizontal $T \cos \phi = \frac{mv^2}{r}$

$$= \frac{mv^2}{\ell \cos \phi}$$

$$T \cos^2 \phi = \frac{mv^2}{\ell}$$

$$\therefore \frac{T \sin \phi}{T \cos^2 \phi} = \frac{kv^2 - mg}{\frac{mv^2}{\ell}}$$

$$\text{ie } \frac{\sin \phi}{\cos^2 \phi} = \frac{k\ell}{m} - \frac{g\ell}{v^2}$$

Question 15 (c) (ii)

Criteria	Marks
• Provides a correct solution	2
• Obtains a quadratic inequality in $\sin\phi$ and attempts to solve it, or equivalent merit	1

Sample answer:

$$\cos^2 \phi > 0 \text{ so}$$

$$\sin \phi < \frac{\ell k}{m} \cos^2 \phi$$

$$= \frac{\ell k}{m} (1 - \sin^2 \phi)$$

$$\therefore \frac{m}{\ell k} \sin \phi < 1 - \sin^2 \phi$$

 m, ℓ, k all positive

$$\sin^2 \phi + \frac{m}{\ell k} \sin \phi - 1 < 0$$

$$\sin \phi < \frac{-\frac{m}{\ell k} + \sqrt{\frac{m^2}{\ell^2 k^2} + 4}}{2} \quad \left(\text{as } \phi < \frac{\pi}{2} \right)$$

$$= \frac{\frac{-m}{\ell k} + \frac{1}{\ell k} \sqrt{m^2 + 4\ell^2 k^2}}{2}$$

$$\therefore \sin \phi < \frac{-m + \sqrt{m^2 + 4\ell^2 k^2}}{2\ell k}$$

Question 15 (c) (iii)

Criteria	Marks
• Provides a correct solution	2
• Differentiates the given expression with respect to ϕ , or equivalent merit	1

Sample answer:

$$F(\phi) = \frac{\sin \phi}{\cos^2 \phi}$$

$$F'(\phi) = \frac{\cos^3 \phi + 2 \sin^2 \phi \cos \phi}{\cos^4 \phi}$$

$$> 0 \left(\text{as, for } -\frac{\pi}{2} < \phi < \frac{\pi}{2}, \cos^3 \phi > 0, \cos \phi > 0, \sin^2 \phi > 0 \text{ and } \cos^4 \phi > 0 \right)$$

 $\therefore F$ is increasing.**Question 15 (c) (iv)**

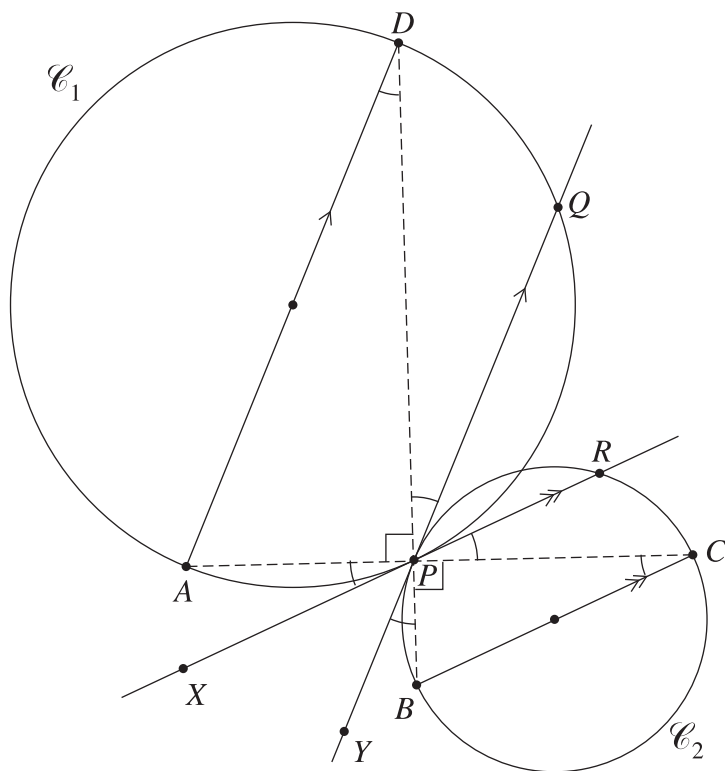
Criteria	Marks
• Provides a correct explanation	1

Sample answer:

As v increases $\frac{\ell g}{v^2}$ decreases, so $\frac{\ell k}{m} - \frac{\ell g}{v^2}$ increases.

That is, $\frac{\sin \phi}{\cos^2 \phi}$ increases.

By (iii) $\frac{\sin \phi}{\cos^2 \phi}$ is an increasing function and so ϕ increases as v increases.

Question 16 (a)**Question 16 (a) (i)**

Criteria	Marks
• Provides a correct solution	2
• Uses angles on alternate segment, or equivalent merit	1

Sample answer:

$\angle APX = \angle ADP$ (The $\angle$ between a tangent and a chord equals the angle in the alternate segment).

$\angle ADP = \angle DPQ$ (Alternate $\angle$ s, $AD \parallel PQ$)

$\therefore \angle APX = \angle DPQ$

Question 16 (a) (ii)

Criteria	Marks
• Provides a correct solution	3
• Makes significant progress towards the solution	2
• Observes result similar to that in part (i) in the other circle, or uses angle in semicircle, or equivalent merit	1

Sample answer:

Using a similar argument to that in part (i), it can be shown that

$$\angle BPY = \angle BCP = \angle CPR$$

$$\angle APD = 90^\circ \text{ (Angle in a semi-circle)}$$

$$\text{Similarly } \angle BPC = 90^\circ$$

$$\angle XPQ = \angle YPR \text{ (Vertically opposite } \angle\text{s)}$$

$$\therefore \angle APX + \angle APD + \angle DPQ = \angle BPY + \angle BPC + \angle CPR$$

$$\therefore 2.\angle APX + 90^\circ = 90^\circ + 2.\angle CPR$$

$$\therefore \angle APX = \angle CPR$$

$$\therefore APC \text{ is a straight line (equal vertically opposite } \angle\text{s)}$$

$$\therefore A, P, C \text{ are collinear.}$$

Question 16 (a) (iii)

Criteria	Marks
• Provides a correct solution	1

Sample answer:

From (i) and (ii)

$\angle APX = \angle CPR$ implies that

$\angle ADP = \angle BCP$

$\therefore A, B, C$ and D are concyclic points (equal $\angle$ s subtended at C and D by interval AB on the same side)

$\therefore ABCD$ is a cyclic quadrilateral.

Question 16 (b) (i)

Criteria	Marks
• Provides a correct solution	3
• Correctly sums and simplifies the middle term, or equivalent merit	2
• Applies formula for geometric series, or equivalent merit	1

Sample answer:

$$\frac{1}{1+x^2} - (1 - x^2 + x^4 - x^6 + \dots + (-1)^{n-1} x^{2(n-1)})$$

$$= \frac{1}{1+x^2} - \frac{1 - (-x^2)^n}{1+x^2} = \frac{1 - 1 + (-x^2)^n}{1+x^2}, \text{ using sum of geometric series.}$$

$$= \frac{(-x^2)^n}{1+x^2}$$

$$\text{Since } 1+x^2 \geq 1 \text{ for all } x, -x^{2n} \leq \frac{(-x^2)^n}{1+x^2} \leq x^{2n}$$

We have

$$-x^{2n} \leq \frac{1}{1+x^2} - (1 - x^2 + x^4 \dots + (-1)^{n-1} x^{2(n-1)}) \leq x^{2n}$$

Question 16 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Integrates some terms in the inequality in (i), or equivalent merit	1

Sample answer:

Inequalities are preserved by integration:

$$\pm \int_0^1 x^{2n} dx = \pm \left. \frac{x^{2n+1}}{2n+1} \right|_0^1 = \pm \frac{1}{2n+1}$$

$$\int_0^1 \frac{1}{1+x^2} dx = \tan^{-1} x \Big|_0^1 = \frac{\pi}{4} - 0 = \frac{\pi}{4}$$

$$\int_0^1 1 - x^2 + x^4 \dots + (-1)^{n-1} x^{2n-2} dx$$

$$= \left[x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots + (-1)^{n-1} \frac{x^{2n-1}}{2n-1} \right]_0^1$$

$$= 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots + (-1)^{n-1} \frac{1}{2n-1}$$

$$\text{so } -\frac{1}{2n+1} \leq \frac{\pi}{4} - \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots + (-1)^{n-1} \frac{1}{2n-1} \right) \leq \frac{1}{2n+1}$$

Question 16 (b) (iii)

Criteria	Marks
• Provides a correct explanation	1

Sample answer:

$$\text{Since } \lim_{x \rightarrow \infty} -\frac{1}{2n+1} = \lim_{x \rightarrow \infty} \frac{1}{2n+1} = 0,$$

then as $n \rightarrow \infty$

$$\frac{\pi}{4} - \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots + (-1)^{n-1} \frac{1}{2n-1} \right) \rightarrow 0$$

$$\therefore \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} \dots$$

Question 16 (c)

Criteria	Marks
• Provides a correct solution	3
• Makes substantial progress	2
• Does substitution $u = 1 + \ln x$ or $u = \ln x$, or equivalent merit	1

Sample answer:

$$I = \int \frac{\ln x}{(1 + \ln x)^2} dx$$

Put $t = \ln x$, then $\frac{dt}{dx} = \frac{1}{x} = \frac{1}{e^t}$

$$I = \int \frac{te^t}{(1+t)^2} dt$$

$$= \int \frac{(1+t)e^t}{(1+t)^2} dt - \int \frac{e^t}{(1+t)^2} dt$$

$$= \int \frac{e^t}{1+t} dt - \left[\frac{e^t}{1+t} + \int \frac{e^t}{1+t} dt \right], \text{ by parts,}$$

$$= \frac{e^t}{1+t} + c$$

$$= \frac{x}{1 + \ln x} + c$$

Mathematics Extension 2

2014 HSC Examination Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	7.6	E4
2	1	7.4	E4
3	1	3.1	E4
4	1	2.1, 2.4	E3
5	1	1.7	E6
6	1	5.1	E7
7	1	4.1	E8
8	1	2.3	E3
9	1	8	HE3, E9
10	1	8	E8, HE6

Section II

Question	Marks	Content	Syllabus outcomes
11 (a) (i)	2	2.2	E3
11 (a) (ii)	2	2.2	E3
11 (b)	3	4.1	E8
11 (c)	3	2.5	E3
11 (d)	2	1.1, 1.2	E6
11 (e)	3	5.1	E7
12 (a) (i)	2	1.3	E6
12 (a) (ii)	2	1.5	E6
12 (b) (i)	1	8.0	E2
12 (b) (ii)	2	8.0	E2
12 (c)	3	1.8	E4, E9
12 (d) (i)	1	4.1	E8
12 (d) (ii)	2	4.1	E8
12 (d) (iii)	2	4.1	E8
13 (a)	3	4.1	E8
13 (b)	4	5.1	E7
13 (c) (i)	1	3.2	E4
13 (c) (ii)	3	3.2	E4, E9
13 (c) (iii)	2	3.2	E4
13 (c) (iv)	2	3.2	E4

Question	Marks	Content	Syllabus outcomes
14 (a) (i)	2	7.2	E3
14 (a) (ii)	2	2.1, 7.2	E3
14 (b) (i)	3	3.1	E3, E9
14 (b) (ii)	2	8.0	E2
14 (c) (i)	2	6.2.1	E5
14 (c) (ii)	4	6.2.1	E5
15 (a)	2	8.3	E2
15 (b) (i)	2	2.4, 8.0	E2, E3
15 (b) (ii)	3	2.4, 8.0	E2
15 (c) (i)	3	6.3.3	E5
15 (c) (ii)	2	8.3	E2
15 (c) (iii)	2	8.0	E2
15 (c) (iv)	1	6.3.3, 8.0	E5
16 (a) (i)	2	8.1	PE3, E2, E9
16 (a) (ii)	3	8.1	PE3, E2, E9
16 (a) (iii)	1	8.1	PE3, E2, E9
16 (b) (i)	3	8.3	E2
16 (b) (ii)	2	8.0	E2
16 (b) (iii)	1	8.0	E2
16 (c)	3	4.1	E8